

statement is True.

b) If the columns of A span $\mathbb{R}^n$, then the columns are linearly independent.

Solution: By the Invertible Matrix Theorem and the equivalence of the statements (h) and (e), we know that the given statement is True.

c) If A is an $n \times n$ matrix, then the equation $Ax=b$ has at least one solution for each b in $\mathbb{R}^n$.

Solution: By the Invertible Matrix Theorem, and the equivalence of the statements (g) and (a), we know that a $n \times n$ matrix A is invertible if and only if the equation $Ax=b$ has at least one solution for each b in $\mathbb{R}^n$. By defn. of inverse, we know that any arbitrary matrix A of size $n \times n$ is not invertible. $\therefore$ The given statement is False.

d) If the equation $Ax=0$ has a nontrivial solution, then A has fewer than n pivot positions.

Solution: By the Invertible Matrix Theorem and the equivalence of the statements (d) and (c), we know that the equation $Ax=0$ has a nontrivial solution if and only if A does not have n pivot positions. By defn, a pivot position of a matrix A corresponds to the leading 1 of a row in rref. By condition 4 from pref, each leading entry of a row is 1. Since a row can have at most 1 leading entry, an $n \times n$ matrix A can have at most n pivots. $\therefore$ The given statement is True.

e) If A^T is not invertible, then A is not invertible.

Solution: By the Invertible Matrix Theorem and the equivalence of the statements (f) and (a), we know that A^T is not invertible if and only if A is not invertible. $\therefore$ The given statement is True.

12a) If there is an $n \times n$ matrix D such that $AD=I$, then there is also an $n \times n$ matrix C such that $CA=I$.

Solution: By the Invertible Matrix Theorem and the equivalence of the statements j and k, we know that the given statement is True.

b) If the columns of A are linearly independent, then the columns of A span $\mathbb{R}^n$.